

GAURAV KUMAR

Lead Cloud / Systems Engineer — Rust, Go, Confidential Computing, Distributed Systems

☎ (+91)-8756746698 | gauravsingh12341@gmail.com | [Website](#) | [GitHub](#) | [LinkedIn](#)

Education

MAY 2019	B. Tech. Mechanical Engineering, IIT Kanpur	7.6 / 10.0
APRIL 2015	Class XII (CBSE), Delhi Public School, Gaya	94.8%

Technical Skills

Languages	Rust (6y), Go (3y), C++ (3y), Python (2y), C (2y)
AI / ML	RAG, Prompt Engineering, Vector Databases, Embeddings, MCP
Containerization	Docker, Kubernetes, OpenShift, Helm, Operators, Docker Compose
Cloud Platforms	Azure, AWS, GCP, IBM Cloud
Databases	Cassandra, CockroachDB, Qdrant, Redis
Debugging	gdb, strace
CI / Automation	Jenkins
Observability	Grafana, Splunk

Work Experience

Lead Cloud Development Engineer — Citrix

July 2025 – Present

- Enhanced Edge Service security by integrating Web Application Firewall (WAF) and rate limiting controls.
- Extended EDT NAK statistics to support range-based reporting.
- Designed and implemented a safe, idiomatic `cgo` wrapper over the EDT C library, enabling production-grade Go APIs.
- Added EDT support in the CGS tool to enable secure, low-latency UDP communication for virtual apps and desktops.

Senior Software Engineer — Fortanix Inc.

June 2019 – June 2025

Armet AI Team

- Designed and developed a confidential Retrieval-Augmented Generation (RAG) platform.
- Implemented core backend services and integrated multiple microservices into a production-grade system.
- Designed and implemented attestation mechanisms for AMD Confidential VMs with GPU support.
- Architected and implemented security hardening for the Qdrant vector database.
- Implemented protections against prompt injection, jailbreak attacks, toxicity, PII exposure, and hallucinations.
- Packaged and deployed Armet AI services in customer-managed VPC environments.

Confidential Computing Manager (CCM) Team

- Conceived and implemented a confidential data clean room.
- Led the Multi-Party Support feature, enabling collaborative workflows across multiple accounts.
- Developed a workflow monitoring microservice for large-scale workflow management.
- Played a key role in launching the CCM SaaS platform (ccm.fortanix.com).
- Architected core backend systems including REST APIs, metering, and audit logging services.
- Delivered advanced features including DCAP attestation, multi-registry support, Node Agent packaging, and Kubernetes Operator integration.
- Enabled secure cluster join using DCAP attestation.
- Migrated CCM services to Ubuntu 24.04.

Data Security Manager

- Designed and implemented a Kubernetes plugin to encrypt Secrets and ConfigMaps in etcd using DSM.
- Built a secure mechanism to inject secrets directly into Kubernetes workloads at runtime.